Two zero-spare command compositions at 215 states

Finite removal of unsafe bootstrap data and protection of an unsafe tail

Research contribution for Vladimir Reshetnikov's ProveIt project

21 October 2026 UTC

Abstract

We construct two materially different binary, single-tape, blank-start machines with 215 working states. Neither is a 214-state result. Both combine the recently published zero-valued spare allocator with a different command language and a new initialization proof. The numerical calculus and its ZFC calibration are explicitly inherited.

The first machine uses command order NIP/MP/DROP. Its first complete source frame is *unsafe*: finite initialization temporarily places the falsity code in both active theorem registers. Rather than asserting an invalid invariant there, we verify a further finite prefix. Failed numerical detachment removes the unsafe data, and after 570,263 binary transitions the machine reaches an ordinary safe root with next-candidate index 5 and additive counter 1. An arbitrary-counter calibration theorem then applies.

The second uses the three-command transfer-pop language. After 73,172 transitions it reaches a rigorously matched interior AX frame. Completing that AX leaves eight theorem heads above a generalization code not established to be a theorem. We do not call that code a theorem: an unbounded credit theorem bounds all future stack debt by three, permanently protecting the arbitrary suffix below at least five theorem heads. The two distinct initialization arguments are accompanied by full arbitrary-natural primitive runtime proofs and exact literal tables. These are ordinary mathematical proofs formalizable in ZFC, with executable certificates, not new Lean/Rocq kernel endpoints or claims of minimum state count.

Contents

1 Exact results and construction boundaries **1**

2 Representation and the zero-spare simulation **3**

3 The numerical calculus and its consistency calibration **3**

4 Primary construction: removing unsafe data during a finite prefix **6**

5 Secondary construction: protecting an unsafe generalization tail **8**

6 Verification, reproduction, and limits **11**

A Complete canonical table: NIP/MP/DROP machine **13**

B Complete canonical table: transfer-pop machine **15**

1 Exact results and construction boundaries

Each machine is deterministic, has one bi-infinite binary tape, and moves left or right at each transition. There are 215 working states, 430 transition rows, and a separate halt label. No

nonblank tape is supplied externally. The claimed conclusion for each literal table M is

$$\mathrm{ZFC} \vdash \big(\mathrm{Halts}(M, \mathrm{blank}) \leftrightarrow \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)\big).$$

Equivalently, blank nonhalting is provably equivalent in ZFC to Con(ZFC). This does not prove actual nonhalting, consistency of ZFC, or a lower bound on the required states.

	NIP/MP/DROP	Transfer-pop
Working states	215	215
Binary transition rows	430	430
Primitive source occurrences	4,751	4,775
Distinct source subtrees	202	201
Guarded primitive cases	6,176	6,209
Symbolic rule applications	1,697,188	1,611,489
Minimum affine duration constant	6	7
Literal prefix used in proof	570,263	73,172
Prefix endpoint	Safe source root	Interior AX frame

Source DAG nodes are not physical states: the second source has 201 nodes but its machine has 215 states. Neither an initialized 214-state runtime nor a rejected 214-state table is substituted for these blank-start machines.

1.1 A small table change with a nontrivial proof obligation

For each construction, begin with its own pinned public 215-state parent, retain its source, change native row 198/0 from $1L40$ to $0L40$, and designate native state 74 as initial rather than 0. All other rows are unchanged. The parent sources differ: the first is the previously uncalibrated order-201 command allocation; the second is the calibrated transfer-pop source. Their parents and exact recipes are included separately.

The changed row makes every source increment append a spare word $B(0) = 01$ rather than $B(1) = 011$, where $B(v) = 0\,1^{v+1}$. From matched frames, the appended spare region after k increments has $2k$ rather than $3k$ cells. This saves k cells in that region, not necessarily k cells of total space at matched binary times. No general running-time improvement is asserted.

A different start state alone would not justify material novelty. Here the physical backend changes, and each combination requires a new source-level initialization argument. The first repairs an order that the public packet retained only as research. The second combines a different three-command language with zero spares and an unsafe-cache bootstrap that the previous unit-spare proof did not cover.

The number of one-writing rows falls by one relative to each unit-spare parent, excluding state-renaming equivalence even with global tape reflection. Further exact rooted graph comparisons against pinned public alternatives are included. These are bounded comparison claims, not assertions of global priority.

1.2 Pinned tables

Primary native SHA-256:

`ef49b414d3d8a08050654d225d4b6ec2972fc73d1bed2f277b5cfded88bdc101`

Primary canonical SHA-256:

`21218bcf54b3feec8813b8266955f05de4f7f07c9a176ab830199f6747628f67`

Secondary native SHA-256:

`29a6ed4a594d7f974cf267a9e6b665af34c9afd8dd7abbbecf0b538ecab151e0`

Secondary canonical SHA-256:

`60eae1c30d214f71c65a207087dc716ba4cdad2a1a62c51e9e107434a0ceb4f1`

Native tables start at 74 and halt at 254; their canonical tables start at 0 and halt at 215. Exact bijections are supplied. File hashes bind the literal serialization as well as the transition content.

2 Representation and the zero-spare simulation

The logical registers are ordered $(N, T, W, A, D, H, X, Y, L)$: next candidate, current theorem, encoded stack, selector residue, additive counter, theorem cache, two scratches, and proof-list remainder. The physical tape stores a fixed-width binary source address, a padding gap, unary primary registers, and a nonempty finite list of spare registers. The spare values need not be theorem codes and have no logical role.

The inherited symbolic checker is parameterized by the appended spare value. Here that parameter is zero, and the changed append block is rederived against each exact table. The domain still includes every natural primary register tuple and padding gap, arbitrary nonempty finite lists of natural spare values, and the complete zero exterior. In particular, mixed zero/one spare lists reached during bootstrap are permitted.

Unary scan and spare-list induction establish unbounded primitive contracts. Every guarded occurrence has a positive affine binary duration, so repeated source execution corresponds to genuine progress. A finite search cutoff for finding a symbolic derivation is not a bound on the natural registers quantified by a successful derivation. Conversely, a timeout would not prove a contract false. Full successful inventories are supplied separately for both tables.

The logical arguments below are independent of spare values. They nevertheless cannot replace the binary proof: the altered append row, source routing, each guard, and each exact initial configuration all require their own verified bindings.

3 The numerical calculus and its consistency calibration

The present construction uses the dominated-parameter calculus supplied in `K218_Dominated_Rho 8_2026-10-20.zip`. This is a recent input to the repository, distinct from its primary, unchanged-calculus K218 table. Its logical theorem is inherited through the 217-state parent. Both present constructions retain that numerical calculus. Their new contributions are the zero-spare command compositions and exact bootstrap calibration below. This distinction is essential for identifying which argument supplies each part of the result.

3.1 Raw numerical operations

For natural numbers put

$$P(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad I(a,b) = 4P(a,b), \qquad a \to b := I(a,b).$$

The remaining constructors are

$$R(a,b) = 4P(a,b) + 1, \quad Q_a b = 4P(a,b) + 2, \quad E(a,b) = 4P(a,b) + 3.$$

Raw truth is $T = 0$ and the refutation target is $F = 1 = R(0,0)$. Write $NX = I(X,F)$ and $\exists_a X = N(Q_a(NX))$. In particular, $I(0,0) = 0$: zero is simultaneously the designated initial theorem and an implication code. All quantified schema parameters below range over *every natural number*. No test that a number decodes as a conventional formula is introduced.

Let $\mathcal{C}$ be the accepted native calculus, called `C222` in the inherited interfaces and used by K219. It has truth, numerical modus ponens, and theorem-only generalization:

$$\frac{\mathcal{C} \vdash X \quad \mathcal{C} \vdash I(X,Y)}{\mathcal{C} \vdash Y}, \qquad \frac{\mathcal{C} \vdash X}{\mathcal{C} \vdash Q_a X}.$$

Its eight logical rows comprise the following seven retained families and the unrestricted family L displayed below:

$$\begin{aligned}
\rho_1(a,b) &= \exists_a E(a,b),\\
\rho_2(a,b,c) &= Q_a(b \to c) \to (Q_a b \to Q_a c),\\
\rho_3(a,b,c) &= E(a,b) \to (R(c,a) \to R(c,b)),\\
\rho_4(a,b,c) &= Q_a Q_b c \to Q_b Q_a c,\\
\rho_5(a,b,c) &= E(c,a) \to (R(c,b) \to R(a,b)),\\
\rho_6(a,b) &= (\exists_a Q_a b) \to b,\\
G(a,b,c) &= E(c,a) \to \big(E(c,b) \to Q_{R(a,b)} E(a,b)\big),\\
L(A,B,C,U) &= ((C \to U) \to A) \to ((A \to C) \to (B \to C)).
\end{aligned}$$

The accepted set-theoretic information remains in the literal gated seed bundle. To specify this inherited object, define

$$\begin{aligned}
B(b,k,m) &= Q_b(m \to N Q_b(m \to N R(b,k))),\\
S(a,b,k) &= Q_k(R(k,a) \to R(k,b)),\\
C(a,b,k,m) &= Q_b(R(b,k) \to N Q_a(m \to N Q_a(S(a,b,k) \to B(b,k,m)))),\\
J(a,b,c) &= Q_c(C(a,b,c,c) \to N B(b,c,c)),\\
V(a,b,c) &= R(a,b) \to Q_c R(a,b),\\
\mathrm{Bundle}(a,b,c) &= N\big(V(a,b,c) \to J(a,b,c)\big).
\end{aligned}$$

A bundle is an axiom when $c \neq a$ and $c \neq b$. Both occurrences of c as matrix code and private label are retained literally. We neither remove these gates nor replace the bundle by an informal set-theoretic analogue. The source returns truth instead when a gate fails.

The established native calibration supplies finite-refutation transformations between $\mathcal{C} \vdash F$ and ZFC inconsistency. That inherited result is a premise of the present construction. It is supported by the repository's native-calculus developments and their connection to the compact set theory underlying K230; it is not proved anew by a transition-table simulation.

For provenance, the dominated package pins its K219 inputs to ProveIt revision `f8ec6158f3bdc885139ee395ae34fe44c3251b9f` and includes its exact native interfaces and finite logic certificate. Earlier proof-assistant endpoints concern their own exact tables. The native-calculus bridges, dominated theorem and present source/runtime arguments must all be included to reach the new table. No new exact-215 proof-assistant endpoint or internal PA proof object is asserted here.

3.2 Why a dominated fourth parameter is sufficient

Define $\mathcal{D}$ by retaining all of the preceding clauses, except that its $L(A,B,C,U)$ axiom requires

$$U \leq A. \tag{D}$$

This is an ordinary inequality between natural codes, not a syntactic freshness test and not an inequality of semantic truth values.

Theorem 3.1 (Inherited dominated-parameter theorem)**.** *For every natural code n,*

$$\mathcal{D} \vdash n \quad \Longleftrightarrow \quad \mathcal{C} \vdash n.$$

The proof is uniform in arbitrary native codes, including zero and all degenerate constructor coincidences.

The easy direction follows because every dominated instance is already an unrestricted instance. For the converse, the inherited finite certificate derives the entire unrestricted L schema from truth, modus ponens, and dominated instances alone. Here are the arithmetic and proof-theoretic facts that make that certificate meaningful.

First, $P(a, b) \geq a, b$ for all naturals. If $a + b = s \geq 1$, then $s(s + 1)/2 \geq s$, and the case $s = 0$ is immediate. Consequently $I(a, b) \geq a, b$. If a schematic term U is literally a subterm of an implication term A, then $U \leq A$ under every numerical assignment. The same conclusion holds whenever $U = T$. This remains true when some implications evaluate to zero. The finite checker uses precisely these sufficient conditions for all dominated leaves.

Second, every certificate node denotes a universally quantified *numerical* statement. Replacing one of its metavariables by an implication term is therefore universal instantiation. It does not assume a general substitution rule inside the object calculus, and it does not rename the numerical labels of a previously decoded atom. At an MP node, equality of the instantiated antecedent and minor is checked exactly. The sole native normalization is $I(T, T) = T$.

The unrestricted-row certificate has the following exact inventory:

Item	Count
Proof nodes	3,728
Shared term nodes	5,160
Dominated-L leaves	1,260
Truth leaves	1
MP steps with explicit substitutions	55
Direct universal-instantiation steps	410
Direct MP steps	2,002
Maximum checked subterm depth	4

These are finite proof objects, not samples of numerical assignments. The implication-subterm inequality proves each axiom side condition for all assignments; induction through the checked proof proves the schematic result for all assignments.

The certificate first obtains identity, weakening K, Peirce's law, composition, and distribution S:

$$\begin{aligned} &K : p \to (q \to p),\\ &\mathrm{P} : ((p \to q) \to p) \to p,\\ &\mathrm{B} : (p \to q) \to ((q \to r) \to (p \to r)),\\ &S : (p \to (q \to r)) \to ((p \to q) \to (p \to r)). \end{aligned}$$

A key intermediate theorem is

$$(p \to q) \to (((p \to r) \to p) \to q).$$

The successful inherited proof of that theorem adds the dominated context $A = ((z \to U) \to v) \to w$; the fourth argument lies three implication edges inside the first. The rest of the fixed derivation reaches unrestricted L. The complete certificate and its checkers are included with the inherited dependency, so the large intermediate formulas need not be expanded into the body of this article.

For clarity, the final passage from K, S, P to L has a short ordinary explanation. Temporarily assume $H = (C \to U) \to A$ and $J = A \to C$. Under the additional assumption $C \to U$, two MPs give A and then C. The deduction transformation justified by K, S gives $(C \to U) \to C$. Peirce's law at C, U gives C. Weakening gives $B \to C$, and discharging J, H yields exactly $L(A, B, C, U)$. No generalization is used under these temporary assumptions. The supplied certificate expands this deduction argument into explicit schema instances and MP steps.

Replacing each unrestricted-L leaf of a finite $\mathcal{C}$ proof by this fixed derived proof gives a finite $\mathcal{D}$ proof of the same code. All other axioms and both rules are retained literally, proving the theorem. Combining it with the inherited native calibration gives

$$\mathcal{D} \vdash 1 \quad \Longleftrightarrow \quad \text{there is a finite ZFC refutation.} \tag{C1}$$

4 Primary construction: removing unsafe data during a finite prefix

4.1 The command language and the unsafe first root

Let P be Cantor pairing and $I(a, b) = 4P(a, b)$. Write the retained tuple as (t, w, d, h). Order 201 means

$$\mathsf{NIP} : 0, \qquad \mathsf{MP} : 1, \qquad \mathsf{DROP} : 2, \qquad \mathsf{AX}(c; x, y) : c + 3.$$

The non-AX field lists are $[0]$, $[1, x]$, and $[2]$ respectively; AX fields are $[c + 3, y, x]$. The primitive bodies and their all-natural command meanings are inherited. Independent control resolvers check the selector partitions, field arities, and normal returns for this exact source.

NIP tests $h = 1$ *before* popping. If it does not halt and $(x, v) = P^{-1}(w)$, it returns (t, v, d, x). DROP returns $(h, P(t, w), d, h)$. MP(x) puts $y = d + x$ and returns

$$\begin{cases} (y, P(0, w), y, 0), & I(t, y) = h, \\ (0, w, y, 0), & I(t, y) \neq h. \end{cases}$$

AX puts $U = d, A = d + x, B = h + y, C = c$, returns the gated bundle as top, resets $d = h = 0$, and pushes the seven logical rows, the dominated row, $Q_U h$, and the old top. The bound $U \leq A$ is automatic for every natural input.

Exact blank execution first reaches a complete zero-address source frame after 118,411 transitions with

$$(N, T, W, A, D, H, X, Y, L) = (2, 0, 2, 0, 0, 0, 0, 0, 0).$$

This is *not* a theorem-stack seed: $2 = P(0, 1)$ contains truth above an unsafe tail. No consistency invariant is applied at that point.

4.2 Four commands that discharge the unsafe suffix

The exact candidate packing determines the next commands, without assumptions on ZFC. Since $2 = P(0, 1)$ and subsequent positive remainders are decremented before decoding, candidate 2 executes NIP,NIP. Candidate $3 = P(2, 0)$ executes DROP. Candidate $4 = P(1, 1)$ executes MP(1) because $1 = P(1, 0)$ supplies the payload.

Boundary	T	W	D	H
Initial complete frame	0	2	0	0
After first NIP	0	1	0	0
After second NIP	0	0	0	1
After DROP	1	0	0	1
After MP(1)	0	0	1	0

Neither NIP halts: both test cache zero, and the second only *then* loads falsity. DROP has no halt test. The final MP fails because $I(1,1) = 16 \neq 1$, clearing both active formula registers. It leaves only the harmless additive counter value 1.

An independently implemented interval-control interpreter and the arithmetic control resolver agree on all source edges. Both reproduce the 435-primitive finite segment, with root visits after 0, 63, 89, 166, and 435 primitives. No halting primitive occurs. The terminal root is

$$(5, 0, 0, 0, 1, 0, 0, 0, 0).$$

A counterfactual theoremhood assertion during the earlier segment would be wrong. A verified finite nonhalting prefix needs no such invariant; the invariant begins only at its safe endpoint.

4.3 A complete literal binary prefix

The proof does not rely solely on the source interpreter for initialization. Independent native and canonical simulation replays all 570,263 binary transitions from blank tape, never halting, and reaches native state 84, head -9, gap 131, with the displayed safe tuple and 176 natural spare words.

The full sparse tape is reconstructed and compared. Its minimum one-cell is $-10 = h - 1$, and its next one is $186 = h + 64 + 131$. Hence all 64 source-address cells are zero. This is a genuine root, not an interior dispatcher frame. Every primary word, every mixed spare value, and the entire zero exterior are checked.

4.4 Strong proof compilation and arbitrary initial counters

The inherited strong compiler supplies, for each numerical theorem F, a finite named-command block $\mathcal{B}_F$ which from every $(t, w, 0, 0)$ either halts or returns $(F, w, 0, 0)$. It quantifies over arbitrary incoming top and stack. Command renumbering transports it to order 201 by changing wire tags, not arithmetic meanings.

For clarity, its principal blocks can be given explicitly. Put

$$\mathsf{Discard} = \mathsf{MP}(1); \mathsf{AX}(1; 0, 0); \mathsf{NIP}^{10}, \quad J(u) = \begin{cases} [], & u = 0, \\ [\mathsf{MP}(u)], & u > 0. \end{cases}$$

From $D = H = 0$, Discard returns truth with the old stack and zero counter/cache. From a zero top/counter/cache, $J(u)$ sets the counter to u; omitting MP when $u = 0$ avoids its zero-alias successful push. Truth uses Discard. A retained row $R_j(a, b, c)$ uses

$$\mathsf{Discard}; \mathsf{AX}(c; a, b); \mathsf{NIP}^{j}; \mathsf{DROP}; \mathsf{NIP}^{11-j}.$$

The dominated row with $u \leq a$ uses

$$\mathsf{Discard}; J(u); \mathsf{AX}(c; a - u, b); \mathsf{NIP}^{8}; \mathsf{DROP}; \mathsf{NIP}^{3}.$$

The bundle uses $\mathsf{Discard};\mathsf{AX}(c;a,b);\mathsf{NIP}^{10}$ when its gates hold. All generated row entries are nonfalsity implication codes and the generalized cache has tag 2, so these selection suffixes restore the underlying stack exactly.

For generalization use

$$\mathcal{B}_X;\mathsf{DROP};J(i);\mathsf{NIP};\mathsf{AX}(i;0,0);\mathsf{NIP}^{9};\mathsf{DROP};\mathsf{NIP};\mathsf{NIP}.$$

For modus ponens from X and $I(X,Y)$ use

$$\mathcal{B}_X;\mathsf{DROP};\mathcal{B}_{I(X,Y)};\mathsf{NIP};\mathsf{DROP};\mathsf{NIP};\mathsf{MP}(Y);\mathsf{AX}(Y;0,0);\mathsf{NIP}^{10};\mathsf{DROP};\mathsf{NIP};\mathsf{NIP}.$$

The AX reset has equal selector/counter parameter and therefore zero gate; the last selection restores Y and the old stack, or halts earlier on falsity. These exact contracts, including the zero implication alias, prove the compiler by induction on a finite numerical derivation.

Theorem 4.1 (Safe root with arbitrary counter)**.** *At a genuine root with $N = n, D = d$ arbitrary naturals, $A = X = Y = L = 0$, and theorem-valued top, cache, and stack heads, source halting is equivalent to numerical inconsistency.*

Proof. Soundness permits every d: AX uses $U = d \le d + x = A$, and all other commands retain their theoremhood arguments. Only cached falsity can trigger halt.

For completeness, append DROP,NIP,NIP to $\mathcal{B}_1$. Prefix by $\mathsf{AX}(0;1,y)$ with $y \ge 1$. It resets counter/cache from every incoming tuple; its resulting top and stack may be arbitrary because the strong block permits them. If R is the successor-based tail encoding of the fixed refutation suffix, the exact first-candidate number is

$$n_y = P(3, P(y, P(1, R))) \ge y.$$

There is no outer successor on a first candidate. Choose $y \ge \max(1, n)$. Every earlier nonhalting candidate drains, because arithmetic terminates and subsequent positive remainders strictly decrease before tail projection. The scheduler thus reaches a successful candidate unless it halts earlier. No bound on an unknown incoming counter is assumed. □

The verified endpoint has $n = 5, d = 1$ and otherwise zero theorem data, so this theorem applies directly. No protected-tail debt theorem is needed for this primary construction.

5 Secondary construction: protecting an unsafe generalization tail

5.1 The transfer-pop language

The second source has three commands: composite transfer-pop S at tag 0, MP at tag 1, and AX at tags $c + 2$, with field lists $[0]$, $[1, x]$, and $[c + 2, y, x]$. Its MP and AX meanings agree with the preceding source. The actual S is old DROP; old NIP; old NIP. Therefore it halts if *either* the incoming top or cache is 1; otherwise, writing $P^{-1}(w) = (x, v)$, its normal result is

$$S(t, w, d, h) = (h, v, d, x).$$

A cache-only halt description would be incomplete. The imported compiler first analyzes that idealized weaker guard and then lifts it to the actual composite by early-halting refinement. On

safe actual executions, both extra tests concern theorem codes, preserving soundness; in the completeness contract an earlier halt is already success.

The initial-state/append-row change leaves all 4,775 source primitives unchanged. The complete AX subtree comprises 4,565 leaves; an independent comparison verifies both their literal operations and their relative paths against the parent.

5.2 An exact interior source boundary, not an invented root

After 73,172 binary transitions from blank tape, the machine is at native dispatcher 84, head -2, gap zero, with 65 natural spare words. The logical tuple is

$$(0, 6, 0, 0, 1, 1, 10, 1, 0).$$

Its source ordinal is 119, with address `11010011000110110110`. This address is *not* zero. Both native and canonical simulators compare every transition and the whole terminal tape; no claim of a root seed is made at this point.

A separate structured-source interpreter begins at AX entry ordinal 25 with

$$(N, T, W, A, D, H, X, Y, L) = (0, 0, 0, 0, 1, 1, 0, 0, 0)$$

and reaches exactly the observed ordinal 119 and entire register store after 177 primitives. The remaining AX execution is total on these natural inputs. Positive unbounded primitive simulation therefore transports the binary machine through the same remaining source suffix. Literal execution of its potentially enormous output codes is not required.

5.3 Eight safe heads, one deliberately untrusted head

The virtual complete AX has $U = A = B = 1, C = 0$. Its gate is *enabled*, because $0 \neq 1$ on both tests. Its output top is the accepted bundle $K(1, 1, 0)$, and it resets counter/cache to zero. Its stack is

$$[R_1(1, 1, 0), \ldots, R_7(1, 1, 0), L(1, 1, 0, 1), Q_1(1), 0; 0].$$

The first eight heads are theorems: the seven retained rows and the dominated row with $U = A = 1$. The incoming cache 1 has not been established to be a theorem. We therefore do *not* claim that its generalization $Q_1(1)$ is a theorem.

Instead isolate an arbitrary suffix

$$J = P(Q_1(1), P(0, 0)) = P(18, 0) = 171.$$

The represented stack consists of eight explicit theorem heads above J. The top is a theorem bundle and the cache is truth. The next-candidate counter remains $N = 0$ and the proof-list remainder is zero. The following unbounded scheduler argument protects that suffix forever, without any theoremhood assumption on its contents.

5.4 The infinite schedule has conservative debt three

Write $f(n)$ for the right projection of Cantor pairing and $T_s = s(s + 1)/2$. Let $W(n)$ be the finite syntactic command word of a freshly fetched candidate, ignoring possible semantic halts. Put $R(0) = []$ and $R(r) = W(r - 1)$ for $r > 0$. If $P^{-1}(n) = (a, y)$, the exact source field arities give

$$W(n) = \begin{cases} S; R(y), & a = 0, \\ \mathsf{MP}; R(f(y)), & a = 1, \\ \mathsf{AX}; R(f(f(y))), & a \geq 2. \end{cases}$$

These words terminate by strict descent of the recursive index. Each $R(r)$ has at most r commands. Assign conservative stack balances $-1, 0, +10$ to S, MP, AX. A successful MP adds a further theorem zero; ignoring it only weakens the protection bound.

Let $b(n)$ be the balance of $W(n)$, $B(k) = \sum_{n<k} b(n)$, $g(0) = 0$, and $g(r) = b(r-1)$ for $r > 0$. If a sequence q has nonnegative initial partial sums, so does $q(f(0)), q(f(1)), \ldots$: the projection sequence concatenates $[0], [0,1], [0,1,2], \ldots$. The assertion follows by summing complete blocks and one partial block, and can be applied twice.

Use the shifted sequence $q(r) = g(r) + 1$. Its first $k \geq 1$ terms sum to $1 + B(k-1) + (k-1)$. The exceptional candidates are

$$W(0) = S, \qquad W(1) = \mathsf{MP}, \qquad W(2) = SS,$$

so $B(0), B(1), B(2), B(3) = 0, -1, -1, -3$, and $B(k) + k \geq 0$ through $k = 3$.

Induct on the Cantor diagonal $s \geq 2$. Its first $s-1$ candidates start with AX, followed by one MP candidate and one S candidate. For the first k AX candidates the balance is

$$10k + \sum_{y<k} g(f(f(y))) \geq 9k,$$

by the twice-applied prefix lemma. It uses only previously proved inequalities at indices below T_s, so the induction is not circular. Inside AX candidate y, its continuation loses at most y, giving lower bound $9y + 10 - y = 8y + 10$. The following MP candidate loses at most $s - 1$, leaving at least $8(s-1)$. The final S and its continuation lose at most $1 + s$, leaving

$$8(s-1) - 1 - s = 7s - 9 > 0.$$

Every command prefix of the diagonal is therefore nonnegative relative to its start. The diagonal boundary remains at least -3, and $B(k) + k \geq 0$ continues to hold, completing the induction. The global conservative minimum from cutoff zero is exactly -3, attained in the exceptional first three candidates.

Lemma 5.1 (Protection of the unsafe suffix)**.** *Starting with eight theorem heads above J, theorem-valued top/cache, $N = L = 0$, the arbitrary suffix J is never exposed during a nonhalting execution. At least five explicit theorem heads remain above it at every command boundary.*

Proof. Track the explicit wrapper length using the conservative balance. AX adds ten theorem heads once the active top/cache are safe; MP preserves or adds a theorem head. Before an S pop the conservative count is at least one more than its post-pop count, which is at least five. Thus it removes a safe head. Inside composite S, DROP first pushes a temporary theorem top, the first NIP removes that temporary head, and the second removes precisely the safe old head accounted for by the balance. Its two halt tests also inspect only theorems.

The wrapper is an explicitly maintained finite list above a fixed tail, not a claim that its length is recoverable from the encoded integer. This handles $P(0,0) = 0$: the exact identity $P^{-1}(P(x,v)) = (x,v)$ still supports each push/pop. The induction applies to every finite actual prefix; defining a syntactic schedule while ignoring halts does not assume nonhalting. □

Consequently every actual halt certifies numerical falsity as a theorem. This supplies soundness despite the initial untrusted cache and buried code 18.

5.5 Completeness is inherited through a strong arbitrary-stack compiler

The transfer source has a complete named-command compiler, not merely a theoremhood invariant. Its strong contract is the same: from every $(t, w, 0, 0)$, a block for F either halts or returns

$(F, w, 0, 0)$. The parent packet contains the full derivation and its separate composite-guard review.

Two structural macros explain why removing primitive DROP does not lose completeness. Put $D_0 = \mathsf{AX}(0; 0, 0)$. Then

$$\mathsf{PUSH} = D_0; S^9; \mathsf{MP}(0), \qquad \mathsf{POP} = D_0; S^{11}.$$

For the ideal cache-guard transfer, PUSH returns $(0, P(t, w), 0, 0)$ on every input, and POP preserves a nonfalsity top while popping one old head and resetting the counter, or halts when the top is 1. For the actual composite, any additional earlier halt is allowed by the strong compiler contract. If no such halt occurs, the displayed normal outputs remain identical. This early-halting distinction is essential; PUSH is not asserted to be nonhalting on arbitrary unsafe inputs for the actual composite.

Axiom selection, theorem generalization, and numerical detachment are built from these macros with exact ten-entry stack cleanup. A compiled derivation of 1, followed by POP, halts on every retained top/stack after the counter and cache are reset. Prefix by $\mathsf{AX}(0; 1, y)$ as before. Its first-candidate integer is now

$$P(2, P(y, P(1, R))) \geq y,$$

because AX begins at tag 2. These successful candidates are cofinal. Finite draining of earlier candidates proves completeness from cutoff zero. The arbitrary tail J causes no obstruction: the compiler's input contract explicitly permits it, while the independent credit argument proves actual-run soundness.

The exact interior bootstrap, total AX suffix, credit theorem, strong compiler, and unbounded binary simulation therefore establish the blank-start consistency equivalence for the secondary table.

6 Verification, reproduction, and limits

Each claimed result has five separate components:

1. A total literal binary table with 215 working states, exact start and halt conventions, and a checked native/canonical bijection.
2. A precise source and its command-calculus calibration. The first uses a tag-permuted strong compiler; the second uses the transfer compiler and its composite-guard refinement.
3. Complete arbitrary-natural primitive contracts, with inductive scan/list lemmas and positive binary durations. The domains include unsafe natural formula codes, which is necessary for both bootstrap proofs.
4. An exact finite blank prefix checked against the whole tape, not just a few decoded registers. The first continues to a safe zero-address root; the second stops at a matched interior AX frame and uses a separately proved total suffix.
5. An all-natural soundness/completeness argument from the calibrated boundary, connected to the explicitly inherited numerical-calculus/ZFC theorem.

Finite source, command, graph, and startup checks support their stated finite claims. They do not substitute for unbounded primitive or scheduler proofs. The mathematical proof engine is executable, but not a proof-assistant kernel. A fresh rerun of that engine is not a second independently designed proof algorithm. Independent parsers, native/dense tape replay, a separately implemented source-prefix check, and mathematical review of the credit proof are identified as such in the receipts.

The archive includes portable entry points for default reconstruction/binding checks and optional complete fresh runtime derivations, together with literal tables, source recipes, compressed

lossless traces, proof texts, attribution, and SHA-256 manifests. Its README specifies the exact commands and distinguishes checking saved inventories from rederiving the symbolic proofs. Python assertions must be enabled; entry points reject optimized Python rather than silently skipping checks.

6.1 What remains open at 214

The investigated 214-state initialized core still fails blank initialization, including with zero-valued spares. Concrete all-start outward-ray certificates exclude those exact tables and starts, not all possible constructions. Larger source re-encoding and helper-allocation searches also found no certified 214-state blank machine. Timeouts and restricted allocation-model failures are not state lower bounds.

The present deliverable satisfies the alternative criterion of materially different 215-state constructions. It does not rename a failed 214-state table, count source nodes as states, supply a nonblank initial tape, or claim that a source-only compiler already gives a smaller binary machine.

6.2 Provenance

The repository is `https://github.com/VladimirReshetnikov/ProveIt`. The public zero-spare packet and command-family source were introduced in commit `cd596116875`; the transfer-pop packet is the one introduced through commits `7342ae8c96c` and `2c77edead6a`. The new work combines those inputs and proves the distinct bootstrap calibrations. The underlying numerical calculus, dominated-row certificate, arithmetic architecture, command compilers, credit theorem, and runtime induction checker retain their upstream attribution and licenses.

The primary numerical-calculus soundness boundary is reached only after the complete 570,263-step finite prefix. The secondary boundary contains only eight assumed theorem heads, not all ten AX outputs. These qualifications are part of the theorem, not merely implementation notes. Neither the repository's uncompiled endpoint drafts nor a literal-data audit is cited as a new kernel-checked endpoint for these tables.

A Complete canonical table: NIP/MP/DROP machine

The working labels are $0, \ldots, 214$, the initial state is 0, and H is halt label 215 with no outgoing rows. Initially the tape is everywhere zero and the head is at coordinate 0. An entry wDq writes w, moves one cell in direction D, and enters q.

State	Read 0	Read 1	State	Read 0	Read 1
0	1R1	1R65	108	0R123	1R9
1	1R93	1R1	109	0R102	1R124
2	0L2	1L3	110	0R152	1R190
3	0L4	1L3	111	0L214	0L96
4	0L6	1L3	112	0R126	1R105
5	1R7	0L5	113	0R201	1R29
6	0L10	1L3	114	0R167	1R184
7	0R7	0R11	115	0R76	1R137
8	0R85	1R194	116	0R150	1R148
9	0R15	1R14	117	0R162	1R205
10	0L16	1L3	118	0L133	0L2
11	0L17	0R11	119	0R40	1R74
12	0L27	1R138	120	0R63	1R134
13	0R20	1R21	121	0R63	1R135
14	0L90	1L32	122	0R72	0L17
15	0L12	0R23	123	0R15	1R134
16	0L24	1L3	124	0R22	1R136
17	1L2	0L2	125	0R149	1R181
18	0R25	0R33	126	0R139	1L21
19	0R206	1R205	127	0R132	1R183
20	0R29	1R30	128	0R209	1R174
21	0R31	1R32	129	0R76	1R179
22	0R25	0R61	130	0R169	1R95
23	0R34	1R23	131	0R70	1R200
24	0L35	1L3	132	0R13	1R143
25	0L36	0R1	133	0R31	1R71
26	0R37	1R26	134	0R40	1L126
27	0R38	1R39	135	0R155	0L17
28	0R40	0L41	136	0R156	0L17
29	0R42	1R43	137	0R163	1R130
30	0R44	1R45	138	0R208	0L214
31	0R46	1R47	139	0R25	0R49
32	0R22	1R48	140	0R142	1R183
33	0R49	1R33	141	0R12	1R158
34	0R33	1R34	142	0R97	1R161
35	0L51	1L3	143	0R178	1R168
36	1R36	1L2	144	0R13	1R161
37	0R52	1R37	145	0R89	1L168
38	0R18	1R202	146	0R89	1R160
39	0R14	1R54	147	0L154	1L165
40	0L12	0R37	148	0R85	1R89
41	0L5	1L55	149	0R161	1R13
42	0R25	0R26	150	0R84	1R145
43	0R14	0L17	151	0R166	1R170
44	0R57	1R58	152	0L68	1L143
45	0R59	1R60	153	0R150	1R207
46	0R25	0R56	154	0R161	1R116
47	0R62	1R63	155	0R98	1R98
48	0R64	0L17	156	0R50	0R33
49	0R26	1R49	157	0R109	1R27
50	0L36	0R65	158	0R113	1R159
51	0L75	1L3	159	0L191	1R97
52	0R17	1R52	160	0R170	1R182
53	0R149	1R145	161	0R20	1R173

State	Read 0	Read 1	State	Read 0	Read 1
54	0*R*69	1*R*70	162	0*R*177	1*R*167
55	0*L*41	1*L*55	163	0*R*161	1*R*97
56	0*R*23	1*R*56	164	0*R*127	1*R*166
57	0*L*12	0*R*52	165	0*R*98	1*R*164
58	0*R*62	0*L*17	166	0*L*8	1*R*144
59	0*R*71	1*R*72	167	0*L*110	1*R*177
60	0*R*63	1*R*73	168	0*R*148	1*R*176
61	0*R*56	1*R*61	169	0*R*131	1*R*212
62	0*R*50	0*R*26	170	0*R*163	1*R*145
63	0*L*29	1*L*74	171	0*R*192	1*R*167
64	0*L*119	1*L*31	172	0*R*181	1*R*181
65	0*R*0	1*R*65	173	0*R*184	1*R*124
66	0*R*98	1*R*152	174	0*R*147	1*R*167
67	1*R*188	1*L*55	175	0*R*171	1*R*88
68	0*R*146	1*R*147	176	0*R*76	1*R*62
69	0*R*77	1*RH*	177	0*R*147	1*R*89
70	0*L*201	1*R*32	178	0*R*142	1*R*85
71	0*R*42	1*R*64	179	0*R*178	1*R*76
72	0*L*90	1*R*79	180	0*R*146	1*R*175
73	0*R*57	1*L*80	181	0*R*144	1*L*151
74	0*R*81	1*R*82	182	0*R*98	1*R*53
75	0*R*75	1*R*83	183	0*R*170	1*R*53
76	0*L*147	1*L*143	184	0*R*18	1*L*133
77	0*R*86	1*R*87	185	0*R*145	1*R*180
78	0*R*201	1*R*142	186	0*R*77	1*R*118
79	0*R*90	1*L*134	187	0*R*107	1*R*126
80	0*L*109	1*R*58	188	0*L*41	1*L*111
81	0*R*50	0*R*34	189	0*R*174	1*R*161
82	0*R*92	0*L*43	190	0*R*152	1*R*117
83	0*R*94	1*R*157	191	0*R*185	1*R*186
84	0*R*97	1*R*13	192	0*R*196	1*R*100
85	0*R*89	1*R*98	193	0*R*148	1*R*203
86	0*R*213	1*R*22	194	0*R*159	1*R*195
87	0*R*118	1*R*101	195	0*R*178	1*R*189
88	0*R*152	1*R*114	196	0*R*66	1*R*8
89	0*R*103	1*R*104	197	0*R*152	1*R*129
90	0*R*81	1*R*80	198	0*R*166	1*R*191
91	0*R*102	1*R*99	199	0*R*154	1*R*125
92	0*R*106	1*R*107	200	0*R*89	1*R*78
93	0*R*67	0*R*1	201	0*R*91	1*R*89
94	0*R*28	1*R*108	202	0*R*141	0*L*214
95	0*R*186	0*L*214	203	0*R*210	1*R*110
96	1*L*111	1*L*96	204	0*R*152	1*R*172
97	0*R*20	1*R*112	205	0*R*204	1*R*162
98	0*R*62	1*R*62	206	0*R*142	1*R*140
99	0*R*105	1*R*97	207	0*R*199	1*R*19
100	0*R*68	1*R*193	208	0*R*211	1*R*13
101	0*R*106	0*L*214	209	0*R*165	1*R*197
102	0*R*119	1*R*120	210	0*R*128	1*R*153
103	0*R*73	1*R*60	211	0*R*18	1*R*115
104	0*R*59	1*R*121	212	0*R*198	1*R*187
105	0*R*22	1*R*122	213	0*R*46	1*R*101
106	0*R*22	0*L*11	214	0*L*5	0*L*214
107	0*R*63	1*R*43			

B Complete canonical table: transfer-pop machine

The same conventions apply, with its own bijection from native labels. These tables are different machines, not two relabelings of one table.

State	Read 0	Read 1	State	Read 0	Read 1
0	1R1	1R64	108	0R148	1R183
1	1R91	1R1	109	0L204	0L94
2	0L2	1L3	110	0R123	1R103
3	0L4	1L3	111	0R206	1R180
4	0L6	1L3	112	0R162	1R178
5	1R7	0L5	113	0R146	1R144
6	0L10	1L3	114	0R157	1R197
7	0R7	0R11	115	0L130	0L2
8	0R83	1R187	116	0R39	1R73
9	0R15	1R14	117	0R62	1R131
10	0L16	1L3	118	0R62	1R132
11	0L17	0R11	119	0R71	0L17
12	0L27	1R128	120	0R15	1R131
13	0R20	1R21	121	0R22	1R133
14	0L88	1L32	122	0R145	1R175
15	0L12	0R23	123	0R135	1L21
16	0L24	1L3	124	0R129	1R177
17	1L2	0L2	125	0R201	1R168
18	0R25	0R33	126	0R75	1R173
19	0R198	1R197	127	0R18	1R174
20	0R29	1R30	128	0R75	1R134
21	0R31	1R32	129	0R13	1R139
22	0R25	0R60	130	0R31	1R70
23	0R34	1R23	131	0R39	1L123
24	0L35	1L3	132	0R151	0L17
25	0L36	0R1	133	0R152	0L17
26	0R37	1R26	134	0R158	1R137
27	0R127	1R38	135	0R25	0R48
28	0R39	0L40	136	0R138	1R177
29	0R41	1R42	137	0R191	1R93
30	0R43	1R44	138	0R95	1R156
31	0R45	1R46	139	0R172	1R163
32	0R22	1R47	140	0R13	1R156
33	0R48	1R33	141	0R87	1L163
34	0R33	1R34	142	0R87	1R155
35	0L50	1L3	143	0L150	1L160
36	1R36	1L2	144	0R83	1R87
37	0R51	1R37	145	0R156	1R13
38	0R14	1R53	146	0R82	1R141
39	0L12	0R37	147	0R161	1R165
40	0L5	1L54	148	0L67	1L139
41	0R25	0R26	149	0R146	1R199
42	0R14	0L17	150	0R156	1R113
43	0R56	1R57	151	0R96	1R96
44	0R58	1R59	152	0R49	0R33
45	0R25	0R55	153	0R69	1R200
46	0R61	1R62	154	0L207	1R95
47	0R63	0L17	155	0R165	1R176
48	0R26	1R48	156	0R20	1R167
49	0L36	0R64	157	0R170	1R162
50	0L74	1L3	158	0R156	1R95
51	0R17	1R51	159	0R124	1R161
52	0R145	1R141	160	0R96	1R159
53	0R68	1R69	161	0L8	1R140
54	0L40	1L54	162	0L108	1R170

State	Read 0	Read 1	State	Read 0	Read 1
55	0R23	1R55	163	0R144	1R169
56	0L12	0R51	164	0R193	1R29
57	0R61	0L17	165	0R158	1R141
58	0R70	1R71	166	0R175	1R175
59	0R62	1R72	167	0R178	1R121
60	0R55	1R60	168	0R143	1R162
61	0R49	0R26	169	0R75	1R61
62	0L29	1L73	170	0R143	1R87
63	0L116	1L31	171	0R103	1R95
64	0R0	1R64	172	0R138	1R83
65	0R96	1R148	173	0R172	1R75
66	1R181	1L54	174	0R184	0L204
67	0R142	1R143	175	0R140	1L147
68	0R76	1RH	176	0R96	1R52
69	0L193	1R32	177	0R165	1R52
70	0R41	1R63	178	0R18	1L130
71	0L88	1R77	179	0R76	1R115
72	0R56	1L78	180	0R105	1R123
73	0R79	1R80	181	0L40	1L109
74	0R74	1R81	182	0R168	1R156
75	0L143	1L139	183	0R148	1R114
76	0R84	1R85	184	0R12	1R97
77	0R88	1L131	185	0R189	1R98
78	0L107	1R57	186	0R144	1R195
79	0R49	0R34	187	0R154	1R188
80	0R90	0L42	188	0R172	1R182
81	0R92	1R194	189	0R65	1R8
82	0R95	1R13	190	0R148	1R126
83	0R87	1R96	191	0R153	1R111
84	0R203	1R22	192	0R150	1R122
85	0R115	1R99	193	0R89	1R87
86	0R148	1R112	194	0R107	1R214
87	0R101	1R102	195	0R202	1R108
88	0R79	1R78	196	0R148	1R166
89	0R100	1R171	197	0R196	1R157
90	0R104	1R105	198	0R138	1R136
91	0R66	0R1	199	0R192	1R19
92	0R28	1R106	200	0R87	1R205
93	0R179	0L204	201	0R160	1R190
94	1L109	1L94	202	0R125	1R149
95	0R20	1R110	203	0R45	1R99
96	0R61	1R61	204	0L5	0L204
97	0R164	1R154	205	0R193	1R138
98	0R67	1R186	206	0R161	1R207
99	0R104	0L204	207	0R208	1R179
100	0R116	1R117	208	0R141	1R209
101	0R72	1R59	209	0R142	1R210
102	0R58	1R118	210	0R211	1R86
103	0R22	1R119	211	0R185	1R162
104	0R22	0L11	212	0R213	1R38
105	0R62	1R42	213	0R13	1R38
106	0R120	1R9	214	0R127	1R212
107	0R100	1R121			